

NAME: _____

TEACHER: _____



The Scots College

Year 12 Mathematics Extension 2

Assessment 1

February 2011

General Instructions

- Working time - 45 minutes
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question

TOTAL MARKS: 36

WEIGHTING: 10 %

Question	Topic	Max Marks	Marks Obtained
1	Complex Numbers	18	
2	Graphs	18	
Total		36	

Question 1 (18 Marks)

- a) (i) Express $\frac{1+i}{1-i}$ in modulus-argument form. [2]
- (ii) Hence, or otherwise, find the value of $\left(\frac{1+i}{1-i}\right)^8$. [1]
- b) Find the square roots of $(-7 + 24i)$ in the form $a + ib$, where $a, b \in R$. [3]
- c) Let $0, z_1$ and z_2 be complex numbers represented in the Argand diagram by O, P and Q respectively. If the point $\frac{z_1}{z_2} \neq 0$, lies on the imaginary axis, show that $OP \perp OQ$. [2]
- d) (i) Indicate the region in the Argand diagram that satisfies simultaneously [3]
 $|z - (1 + i)| \leq 1$ and $\frac{\pi}{4} \leq \arg([z - (1 + i)]) \leq \frac{\pi}{2}$.
- (ii) Find the maximum value of $|z|$ in the region mentioned in (i) and also find the specific complex number z that corresponds to this maximum value. [2]
- e) Let a complex number be given by $z = \cos \theta + i \sin \theta$.
- (i) Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ and $z^n - \frac{1}{z^n} = 2i \sin n\theta$. [1]
- (ii) By factorizing $z^6 + 1$, express $z^6 + 1$ as a product of three **quadratic** factors of z . [2]
- (iii) By using the results of (i) and (ii), or otherwise, show that [2]
 $\cos 3\theta = \cos \theta (2 \cos \theta - \sqrt{3})(2 \cos \theta + \sqrt{3})$

.....Question 2 on next page

Question 2 (18 Marks)

a) Consider the function $y = \frac{1+x^2}{x}$.

(i) Find the stationary points and points of inflexion, if any, for the function. [2]

(ii) Find the equation(s) of any asymptote(s). [1]

(iii) Sketch the graph of $y = f(x)$, showing clearly any intercepts, stationary points and points of inflexion, if they exist, and equation of any asymptote. [3]

(iv) Hence, sketch, on the same set of axis as in (iii), the graph of $y = \frac{x}{1+x^2}$, indicating clearly the intercepts, stationary points and asymptotes. [2]

b) Consider the curve $2x^2 + xy - y^2 = 0$. Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at the point (2,4) on the curve. [4]

c) The graph given below (see next page) represents the function $y = 1 - 2e^{-x^2}$. The line $y = 1$ is the horizontal asymptote. In the graphs provided in the answer booklet, sketch the function represented by

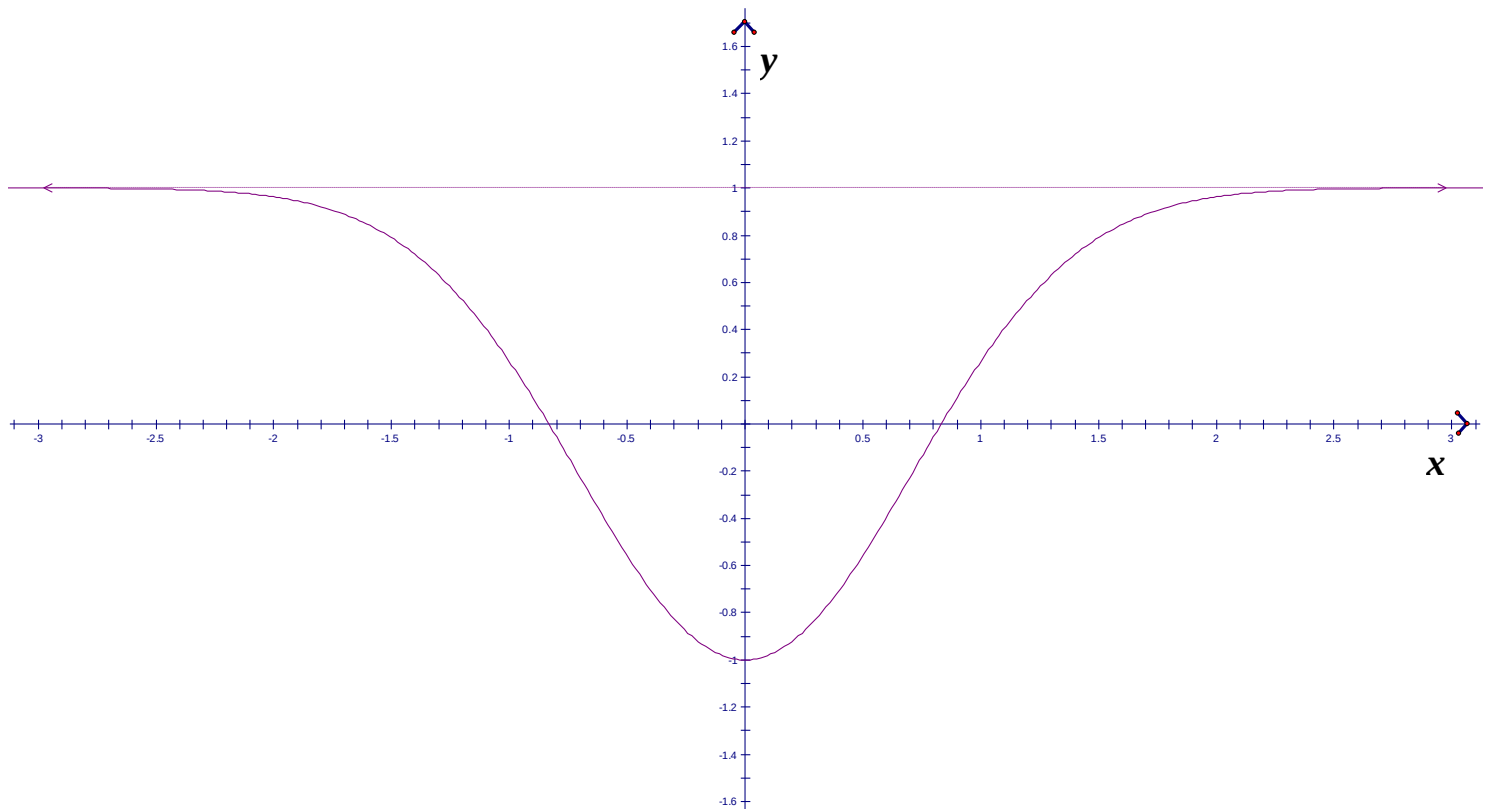
(i) $y = [f(x)]^2$ [2]

(ii) $y^2 = f(x)$ [2]

(iii) $|y| = f(x)$ [2]

.....Question 2 continued on next page

Question 2 c) : Graph of $y = 1 - 2e^{-x^2}$.



... END OF EXAMINATION ...



Mathematics Extension 2. Assessment 1.

ANSWER SHEET

Feb 2011

Name: SOLUTIONS

Teacher: _____

Question No. 1.

$$\begin{aligned} (2) \quad (i) \quad & \frac{1+i}{1-i} \\ &= \frac{\sqrt{2} \operatorname{cis} \pi/4}{\sqrt{2} \operatorname{cis} (-\pi/4)} \checkmark \\ &= \operatorname{cis} \left(\frac{\pi}{4} + \frac{\pi}{4} \right) \\ &= \operatorname{cis} \pi/2 \\ &= \underline{\underline{\cos \pi/2 + i \sin \pi/2}} \checkmark \end{aligned}$$

$$(ii) \quad \operatorname{cis} \pi/2 = i$$

$$\therefore \frac{1+i}{1-i} = i$$

$$\left(\frac{1+i}{1-i} \right)^8 = i^8 = \underline{\underline{1}} \checkmark$$

$$(b) \quad \text{let } \sqrt{-7+24i} = a+ib$$

$$-7+24i = a^2 + 2iab - b^2$$

$$a^2 - b^2 = -7 \quad \text{--- (i)}$$

$$2ab = 24 \Rightarrow ab = 12 \quad \text{--- (ii)} \Rightarrow b = 12/a \quad \checkmark$$

$$a^2 - \frac{144}{a^2} = -7 \Rightarrow a^4 + 7a^2 - 144 = 0$$
$$(a^2 + 16)(a^2 - 9) = 0$$

Q 1 (b)

$$(a^2 + 16)(a^2 - 9) = 0$$

$$\therefore a^2 = -16 \quad \text{or} \quad a^2 = 9$$

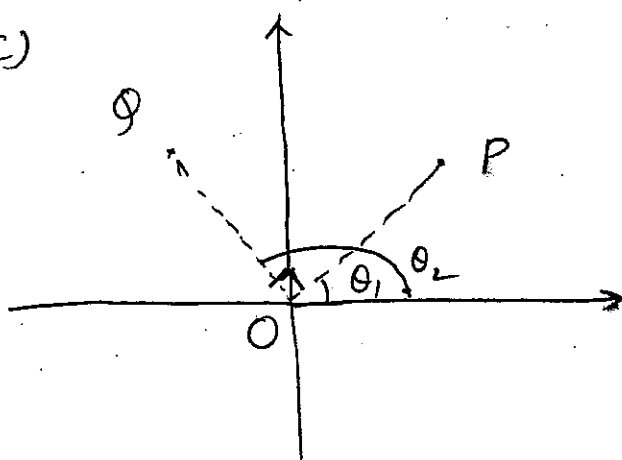
$$(\text{re}' \because a \in \mathbb{R}) \quad a = \pm 3 \quad \checkmark$$

$$\text{when } a = 3, \quad b = 4$$

$$a = -3 \quad b = -4$$

$\therefore$ square roots of $-7 + 24i$ are $\pm (3 + 4i)$ $\checkmark$

(c)



if $\frac{z_2}{z_1}$ lies on the imaginary axis then

$$\frac{z_2}{z_1} = ki \quad \checkmark \quad \text{where } k \text{ is a real no.}$$

$$\therefore z_2 = ki z_1 = k \cos \frac{\pi}{2} \cdot z_1$$

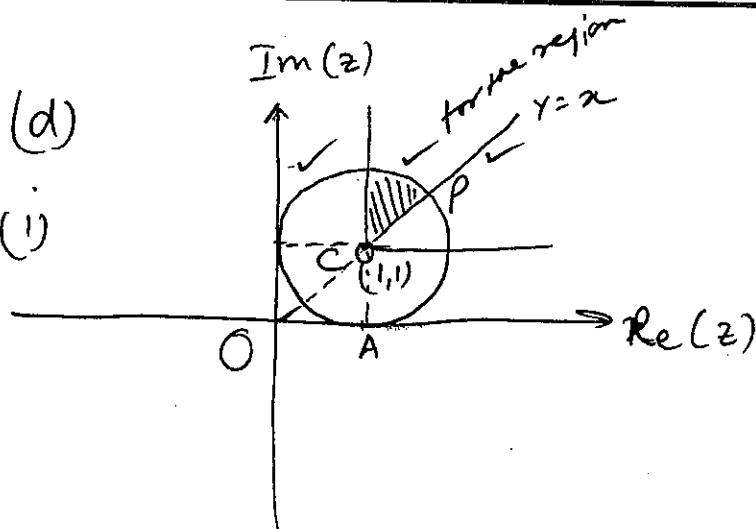
$$\arg(z_2) = \arg z_1 + \frac{\pi}{2}$$

$$\therefore \angle POQ = 90^\circ \quad \checkmark$$

$$\text{or } \underline{\underline{OP \perp OQ}}$$

Q 1 (d)

(i)



(ii) The maximum value of $|z| = OP$ which corresponds to the complex no. representing P (lying on the line $y = x$).

$$CP = 1 \quad OC = \sqrt{OA^2 + AC^2} \\ = \sqrt{1+1} = \sqrt{2}$$

$$\therefore OP = 1 + \sqrt{2}$$

$$\therefore \text{Max. value of } |z| = 1 + \sqrt{2} \quad \checkmark$$

$$P \text{ is representing } \underline{\underline{z = (1 + \sqrt{2}) \operatorname{cis} \pi/4}}$$

$$\text{or } z = \frac{1 + \sqrt{2}}{\sqrt{2}} + i \frac{1 + \sqrt{2}}{\sqrt{2}} \quad \checkmark$$

$$z = \frac{\sqrt{2} + 2}{2} + \frac{\sqrt{2} + 2}{2} i$$

$$(e) (i) z = \cos \theta + i \sin \theta$$

$$z^n = \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i \sin(-n\theta) \\ = \cos n\theta - i \sin n\theta$$

$$\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta, \quad z^n - \frac{1}{z^n} = 2i \sin n\theta$$

$$(ii) \quad z^6 + 1 = (z^2)^3 + 1^3 \\ = (z^2 + 1)(z^4 - z^2 + 1) \\ = (z^2 + 1)(z^2 + 1)^2 - 3z^2 \\ = (z^2 + 1)(z^2 + 1 - \sqrt{3}z)(z^2 + 1 + \sqrt{3}z)$$

$$(iii) \quad z^6 + 1 = (z^2 + 1)(z^2 + 1 - \sqrt{3}z)(z^2 + 1 + \sqrt{3}z) \div z^3$$

$$z^3 + \frac{1}{z^3} = \left(z + \frac{1}{z}\right) \left(z + \frac{1}{z} - \sqrt{3}\right) \left(z + \frac{1}{z} + \sqrt{3}\right)$$

or $2 \cos 3\theta = 2 \cos \theta (2 \cos \theta - \sqrt{3})(2 \cos \theta + \sqrt{3})$
(from the results of (i))

$$\cos 3\theta = \cos \theta (2 \cos \theta - \sqrt{3})(2 \cos \theta + \sqrt{3})$$



ANSWER SHEET

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Question No. 2

$$(a) \quad y = \frac{1+x^2}{x}$$
$$= \frac{1}{x} + x$$

$$(i) \quad \frac{dy}{dx} = -\frac{1}{x^2} + 1 \quad \frac{d^2y}{dx^2} = \frac{2}{x^3}$$

for st. pt.:

$$\frac{dy}{dx} = 0 \quad -\frac{1}{x^2} + 1 = 0$$

$$\frac{1}{x^2} = 1$$

$$x^2 = 1$$

$$x = 1, y = \frac{1+1}{1} = 2$$

$$\text{or } x = -1, y = \frac{1+1}{-1} = -2$$

When $x = 1$, $\frac{d^2y}{dx^2} = \frac{2}{1} > 0 \therefore (1, 2)$ is a main pt. ✓

$x = -1$, $\frac{d^2y}{dx^2} = \frac{2}{-1} < 0 \therefore (-1, -2)$ is a min. pt.

$$\text{for poi } \frac{d^2y}{dx^2} = 0 \text{ or } \frac{2}{x^3} = 0$$

no soln. $\therefore$ no poi ✓

Q2 (a)

(ii) $y = \frac{1}{x} + x$

when $x \rightarrow \infty$, $\frac{1}{x} \rightarrow 0$, $y \rightarrow x$

$\therefore y = x$ is the oblique asymptote

$x = 0$ is the vertical asymptote ✓

(iii) $f(x) = \frac{1+x^2}{x}$, $x=0$ y is undefined $\therefore$ no y -int.

$$f(-x) = \frac{1+(-x)^2}{-x} = -\frac{1+x^2}{x}$$

$$f(-x) = -f(x)$$

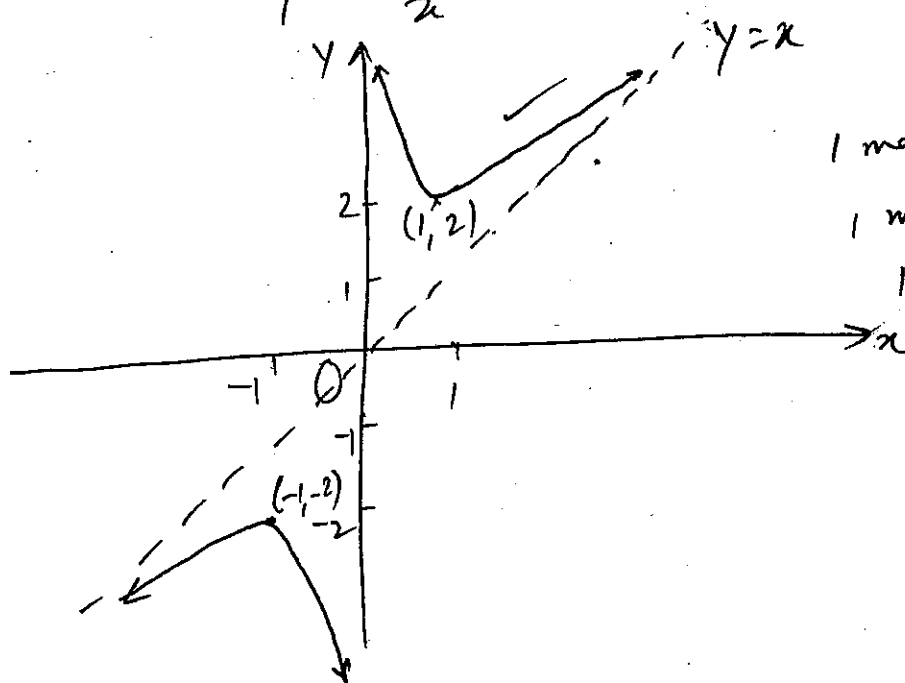
$\therefore y = \frac{1+x^2}{x}$ is an odd function

$$y=0$$

$$1+x^2=0$$

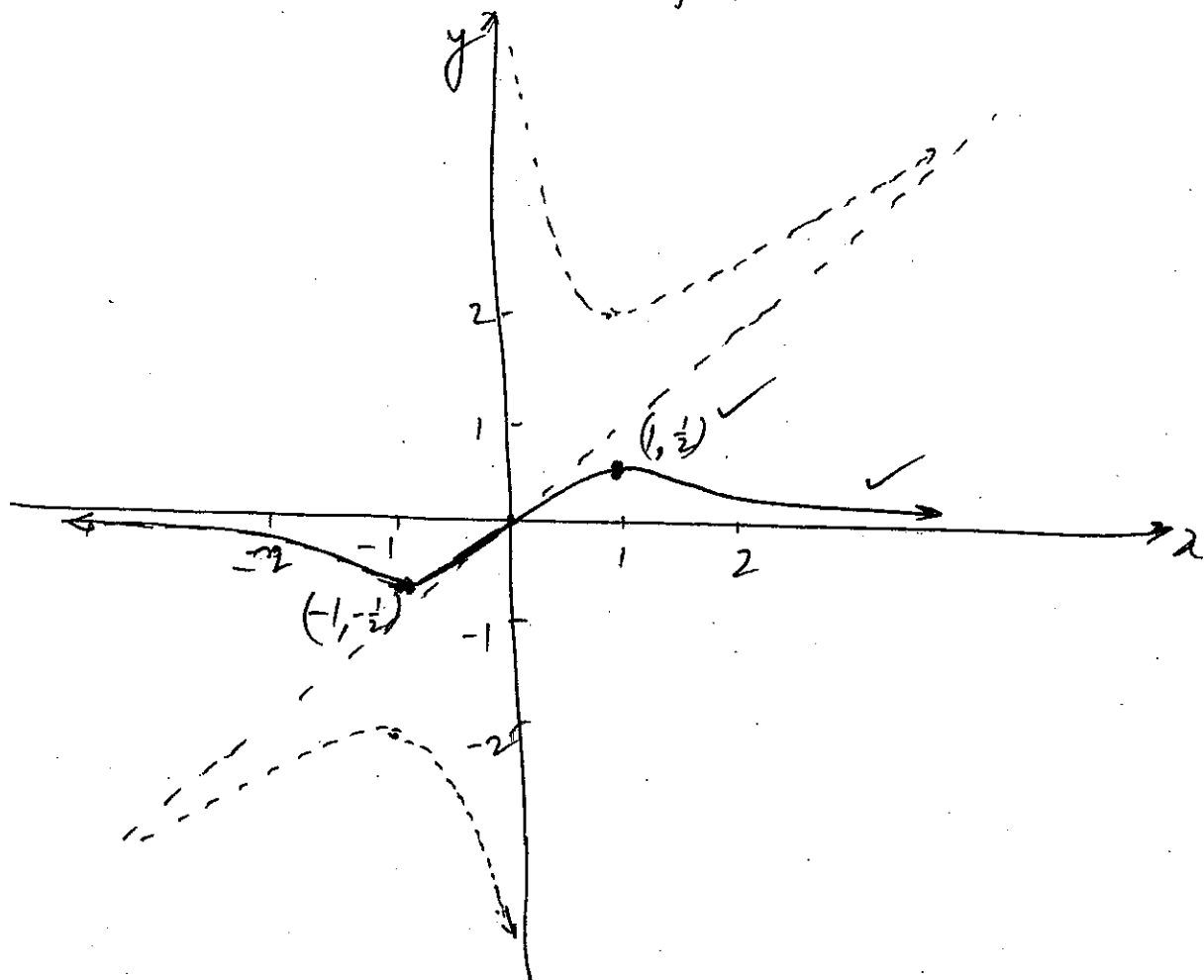
no solution

$\therefore$ no x -int.



1 mark st. pt. & asymptote
1 mark shape
1 mark labels

Q 2(a) (iv) $y = \frac{x}{1+x^2} = \frac{1}{f(x)}$



Q 2 (b)

$$2x^2 + xy - y^2 = 0$$

Differentiating w.r.t. x

$$4x + x \frac{dy}{dx} + y(1) - 2y \frac{dy}{dx} = 0$$

$$\text{Rearranging } 4x + y = (2y - x) \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{4x+y}{2y-x} \quad \checkmark \quad u = 2y-x \quad v = 4x+y$$

$$u' = 2y' - 1 \quad v' = 4 + y'$$

$$\frac{d^2y}{dx^2} = \frac{(2y-x)(4+y') - (4x+y)(2y'-1)}{(2y-x)^2} \quad \checkmark$$

When $x = 2, y = 4$

$$\frac{dy}{dx} = \frac{8+4}{8-2} = \frac{12}{6} = \underline{\underline{2}} \quad \checkmark$$

$$\frac{d^2y}{dx^2} = \frac{(8-2)(4+2) - (8+4)(4-1)}{(8-2)^2}$$

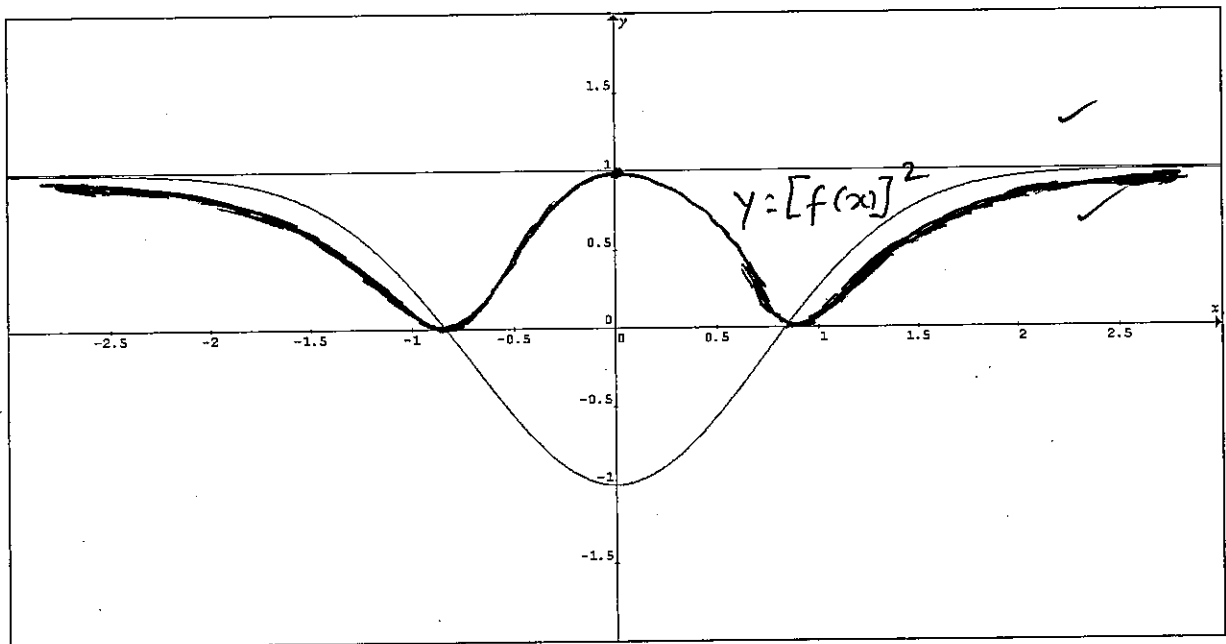
$$= \frac{36 - 36}{36} = -\frac{0}{36}$$

$$= \underline{\underline{0}} \quad \checkmark$$

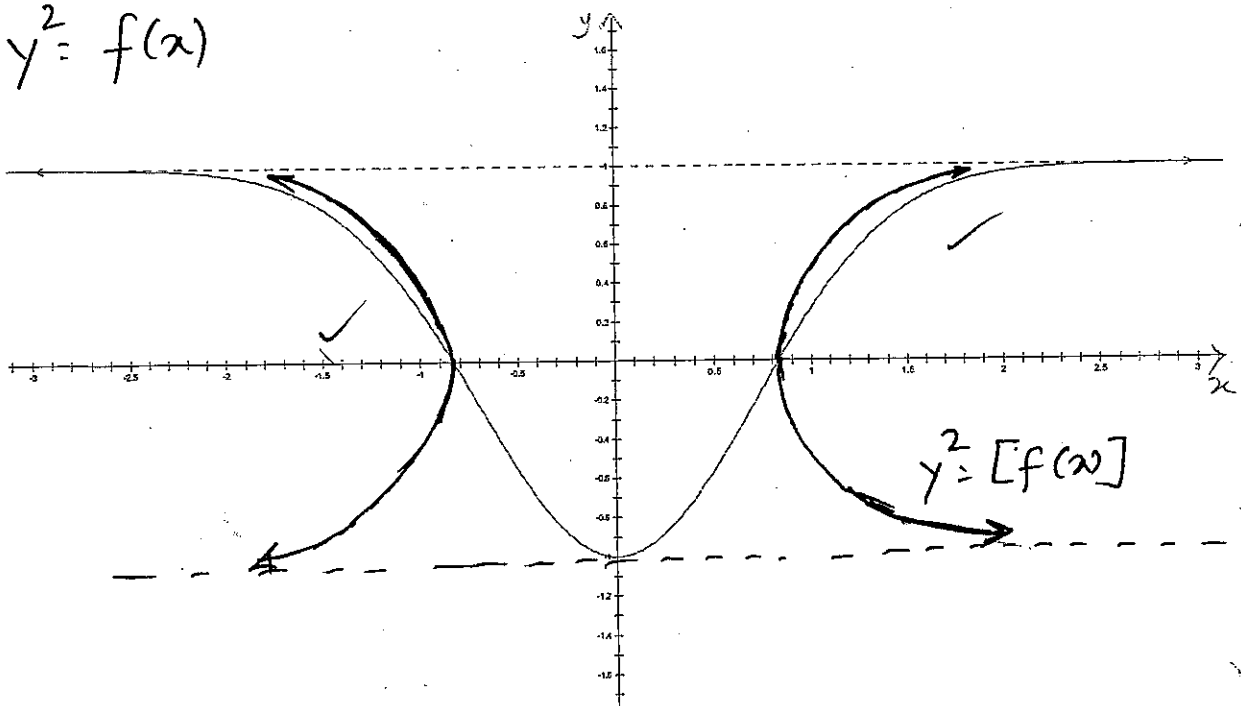
Q2 (a)

(i) $y = [f(x)]^2$

-1 for any mistake
in each graph.



(ii) $y^2 = f(x)$



Q 2(c)

(iii) $|y| = f(x)$

